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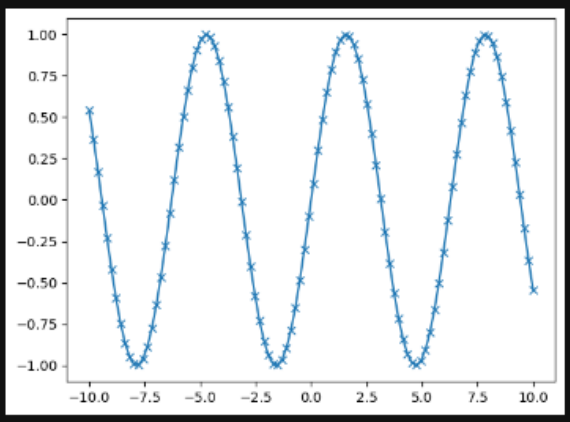
Code JupyterLab Python 3 (ipykernel)

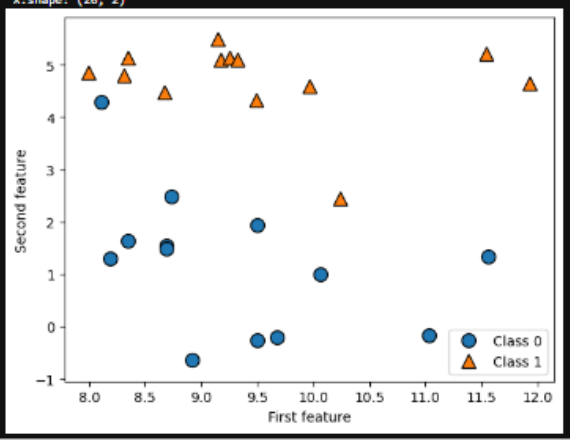
[1]:
```
%matplotlib inline
import numpy as np
import matplotlib.pyplot as plt
# Generate a sequence of numbers from -10 to 10 with 100 steps in between
x = np.linspace(-10, 10, 100)
# Create a second array using sine
y = np.sin(x)
# The plot function makes a line chart of one array against another
plt.plot(x, y, marker="x")
```

[1]:
```
[<matplotlib.lines.Line2D at 0x299ec262fd0>]
```



[4]:
```
import mglearn
import matplotlib.pyplot as plt
# generate dataset
X, y = mglearn.datasets.make_forge()
# plot dataset
mglearn.discrete_scatter(X[:, 0], X[:, 1], y)
plt.legend(["Class 0", "Class 1"], loc=4)
plt.xlabel("First feature")
plt.ylabel("Second feature")
print("X.shape: {}".format(X.shape))
```

```
X.shape: (26, 2)
```



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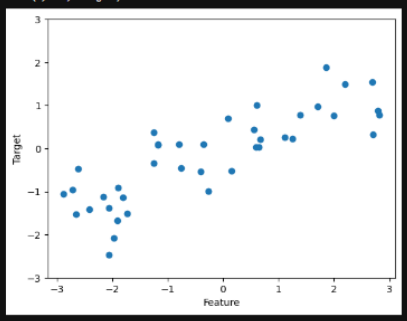
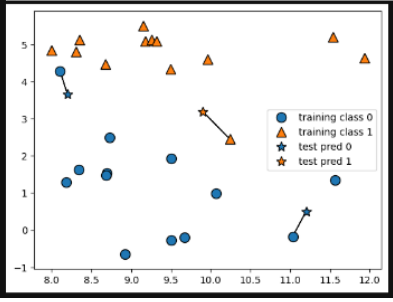
Code JupyterLab Python 3 (ipykernel)

[5]:
```
X, y = mglearn.datasets.make_wave(n_samples=40)
plt.plot(X, y, 'o')
plt.ylim(-3, 3)
plt.xlabel("Feature")
plt.ylabel("Target")
```

[5]: Text(0, 0.5, 'Target')



[8]:
```
mglearn.plots.plot_knn_classification(n_neighbors=1)
# 3, 5, 7, 9
```



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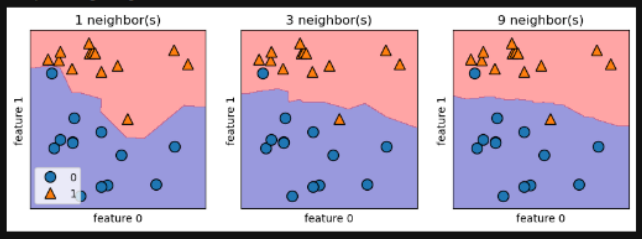
Code JupyterLab Python 3 (ipykernel)

[9]:
```
from sklearn.model_selection import train_test_split
X, y = mglearn.datasets.make_forge()
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)

from sklearn.neighbors import KNeighborsClassifier
clf = KNeighborsClassifier(n_neighbors=3)

clf.fit(X_train, y_train)

print("Test set predictions: {}".format(clf.predict(X_test)))

print("Test set accuracy: {:.2f}".format(clf.score(X_test, y_test)))
```

```
Test set predictions: [1 0 1 0 1 0 0]
Test set accuracy: 0.86
```

[10]:
```
fig, axes = plt.subplots(1, 3, figsize=(10, 3))

for n_neighbors, ax in zip([1, 3, 9], axes):
    # the fit method returns the object self, so we can instantiate
    # and fit in one line
    clf = KNeighborsClassifier(n_neighbors=n_neighbors).fit(X, y)
    mglearn.plots.plot_2d_separator(clf, X, fill=True, eps=0.5, ax=ax, alpha=.4)
    mglearn.discrete_scatter(X[:, 0], X[:, 1], y, ax=ax)
    ax.set_title("{} neighbor(s)".format(n_neighbors))
    ax.set_xlabel("feature 0")
    ax.set_ylabel("feature 1")
axes[0].legend(loc=3)
```

[10]: <matplotlib.legend.Legend at 0x299f2b47ed0>



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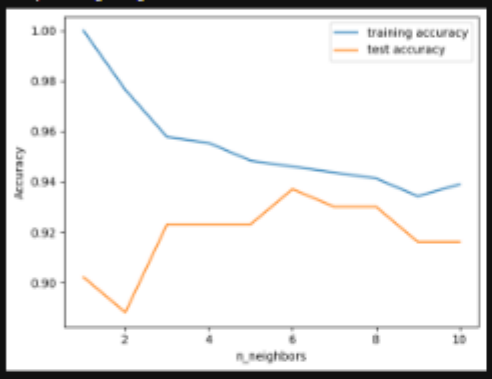
Code JupyterLab Python 3 (ipykernel)

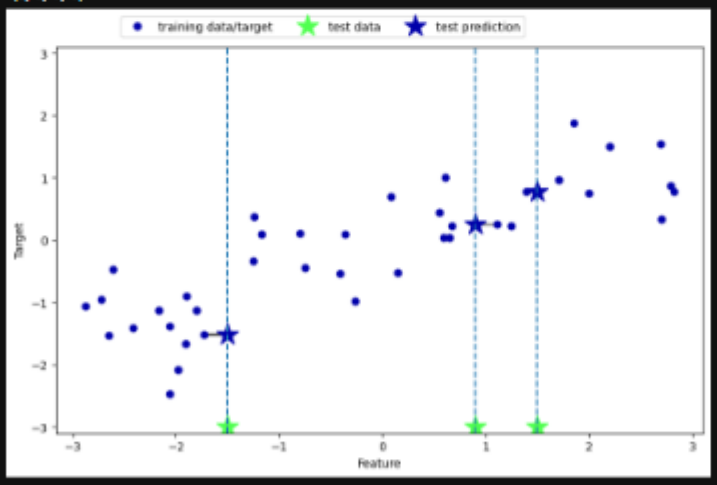
[11]:
```
from sklearn.datasets import load_breast_cancer

cancer = load_breast_cancer()
X_train, X_test, y_train, y_test = train_test_split(
    cancer.data, cancer.target, stratify=cancer.target, random_state=66)

training_accuracy = []
test_accuracy = []
# try n_neighbors from 1 to 10
neighbors_settings = range(1, 11)

for n_neighbors in neighbors_settings:
    # build the model
    clf = KNeighborsClassifier(n_neighbors=n_neighbors)
    clf.fit(X_train, y_train)
    # record training set accuracy
    training_accuracy.append(clf.score(X_train, y_train))
    # record generalization accuracy
    test_accuracy.append(clf.score(X_test, y_test))

plt.plot(neighbors_settings, training_accuracy, label="training accuracy")
plt.plot(neighbors_settings, test_accuracy, label="test accuracy")
plt.ylabel("Accuracy")
plt.xlabel("n_neighbors")
plt.legend()
```

[11]:
```
<matplotlib.legend.Legend at 0x209f2066710>
```



[14]:
```
mglearn.plots.plot_knn_regression(n_neighbors=3)
```

[14]:
```
(4, 5, 7, 9)
```



[15]:
```
from sklearn.neighbors import KNeighborsRegressor

X, y = mglearn.datasets.make_wave(n_samples=40)

# split the wave dataset into a training and a test set
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)

# instantiate the model and set the number of neighbors to consider to 3
reg = KNeighborsRegressor(n_neighbors=3)
# fit the model using the training data and training targets
reg.fit(X_train, y_train)

print("Test set predictions:\n{}".format(reg.predict(X_test)))

print("Test set R^2: {:.2f}".format(reg.score(X_test, y_test)))
```

```
Test set predictions:
[-0.05396539  0.35686046  1.13671923 -1.89415682 -1.13881398 -1.63111882
  0.35686046  0.91771374 -0.44680446 -1.13881398]
Test set R^2: 0.83
```

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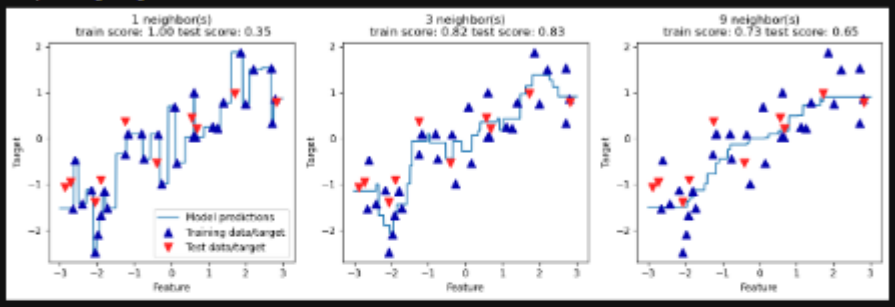
Code JupyterLab Python 3 (ipykernel)

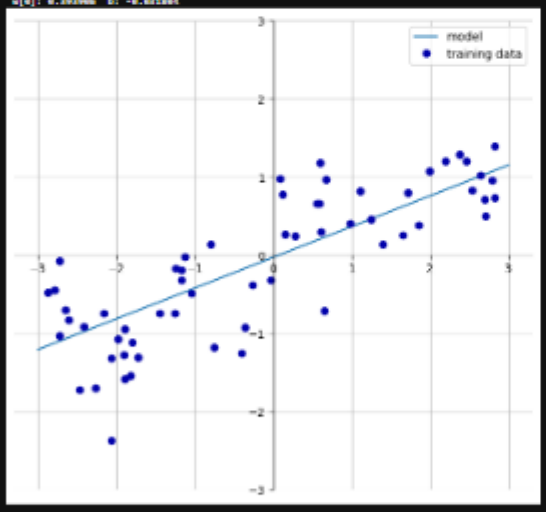
[16]:
```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
# create 1,000 data points, evenly spaced between -3 and 3
line = np.linspace(-3, 3, 1000).reshape(-1, 1)
for n_neighbors, ax in zip([1, 3, 9], axes):
    # make predictions using 1, 3, or 9 neighbors
    reg = KNeighborsRegressor(n_neighbors=n_neighbors)
    reg.fit(X_train, y_train)
    ax.plot(line, reg.predict(line))
    ax.plot(X_train, y_train, '^', c=mglearn.cm2(0), markersize=8)
    ax.plot(X_test, y_test, 'v', c=mglearn.cm2(1), markersize=8)

    ax.set_title("{} neighbor(s)\n train score: {:.2f} test score: {:.2f}".format(
                 n_neighbors, reg.score(X_train, y_train),
                 reg.score(X_test, y_test)))
    ax.set_xlabel("Feature")
    ax.set_ylabel("Target")
axes[0].legend(["Model predictions", "Training data/target",
                "Test data/target"], loc="best")
```

[16]:
```
<matplotlib.legend.Legend at 0x2998182ed60>
```



[17]:
```
mglearn.plots.plot_linear_regression_wave()
```

```
w[0]: 0.393906  b: -0.031804
```



[18]:
```
from sklearn.linear_model import LinearRegression
X, y = mglearn.datasets.make_wave(n_samples=60)
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)

lr = LinearRegression().fit(X_train, y_train)

print("lr.coef_: {}".format(lr.coef_))
print("lr.intercept_: {}".format(lr.intercept_))

print("Training set score: {:.2f}".format(lr.score(X_train, y_train)))
print("Test set score: {:.2f}".format(lr.score(X_test, y_test)))
```

```
lr.coef_: [0.39390555]
lr.intercept_: -0.031804343026759746
Training set score: 0.67
Test set score: 0.66
```

[19]:
```
X, y = mglearn.datasets.load_extended_boston()

X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)
lr = LinearRegression().fit(X_train, y_train)

print("Training set score: {:.2f}".format(lr.score(X_train, y_train)))
print("Test set score: {:.2f}".format(lr.score(X_test, y_test)))
```

```
Training set score: 0.95
Test set score: 0.61
```

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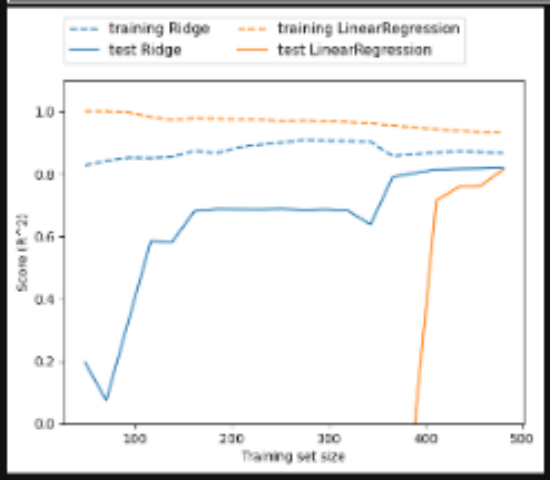
Code JupyterLab Python 3 (ipykernel)

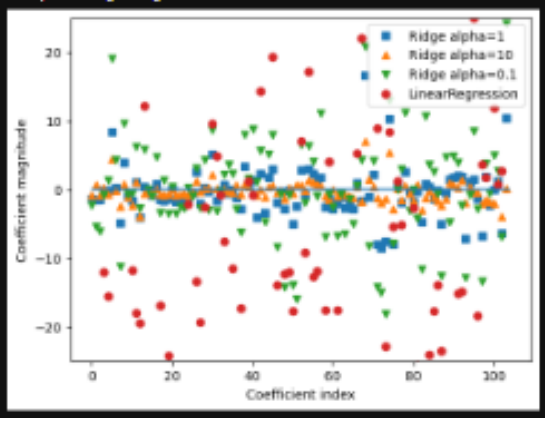
[20]:
```
mglearn.plots.plot_ridge_n_samples()
```



[21]:
```
from sklearn.linear_model import Ridge

#Ridge alpha value of 1
ridge = Ridge().fit(X_train, y_train)
print("Training set score: {:.2f}".format(ridge.score(X_train, y_train)))
print("Test set score: {:.2f}".format(ridge.score(X_test, y_test)))

#Ridge alpha value of 10
ridge10 = Ridge(alpha=10).fit(X_train, y_train)
print("Training set score: {:.2f}".format(ridge10.score(X_train, y_train)))
print("Test set score: {:.2f}".format(ridge10.score(X_test, y_test)))

#Ridge alpha value of 0.1
ridge01 = Ridge(alpha=0.1).fit(X_train, y_train)
print("Training set score: {:.2f}".format(ridge01.score(X_train, y_train)))
print("Test set score: {:.2f}".format(ridge01.score(X_test, y_test)))

#Print alpha value using plt
plt.plot(ridge.coef_, '^', label="Ridge alpha=1")
plt.plot(ridge10.coef_, '*', label="Ridge alpha=10")
plt.plot(ridge01.coef_, 'v', label="Ridge alpha=0.1")
plt.plot(lr.coef_, 'o', label="LinearRegression")
plt.xlabel("Coefficient index")
plt.ylabel("Coefficient magnitude")
plt.hlines(0, 0, len(lr.coef_))
plt.ylim(-25, 25)
plt.legend()
```

```
Training set score: 0.89
Test set score: 0.75
Training set score: 0.79
Test set score: 0.64
Training set score: 0.93
Test set score: 0.77
```

[21]:
```
<matplotlib.legend.Legend at 0x29981970410>
```



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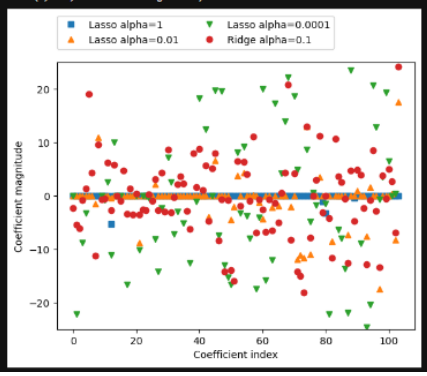
Code JupyterLab Python 3 (ipykernel)

[22]:
```python
from sklearn.linear_model import Lasso
lasso = Lasso().fit(X_train, y_train)
print("Training set score: {:.2f}".format(lasso.score(X_train, y_train)))
print("Test set score: {:.2f}".format(lasso.score(X_test, y_test)))
print("Number of features used: {}".format(np.sum(lasso.coef_ != 0)))

# we increase the default setting of "max_iter",
# otherwise the model would warn us that we should increase max_iter.
lasso001 = Lasso(alpha=0.01, max_iter=100000).fit(X_train, y_train)
print("Training set score: {:.2f}".format(lasso001.score(X_train, y_train)))
print("Test set score: {:.2f}".format(lasso001.score(X_test, y_test)))
print("Number of features used: {}".format(np.sum(lasso001.coef_ != 0)))


lasso00001 = Lasso(alpha=0.0001, max_iter=100000).fit(X_train, y_train)
print("Training set score: {:.2f}".format(lasso00001.score(X_train, y_train)))
print("Test set score: {:.2f}".format(lasso00001.score(X_test, y_test)))
print("Number of features used: {}".format(np.sum(lasso00001.coef_ != 0)))


plt.plot(lasso.coef_, 's', label="Lasso alpha=1")
plt.plot(lasso001.coef_, '^', label="Lasso alpha=0.01")
plt.plot(lasso00001.coef_, 'v', label="Lasso alpha=0.0001")
plt.plot(ridge01.coef_, 'o', label="Ridge alpha=0.1")
plt.legend(ncol=2, loc=(0, 1.05))
plt.ylim(-25, 25)
plt.xlabel("Coefficient index")
plt.ylabel("Coefficient magnitude")
```

```
Training set score: 0.29
Test set score: 0.21
Number of features used: 4
Training set score: 0.90
Test set score: 0.77
Number of features used: 33
Training set score: 0.95
Test set score: 0.64
Number of features used: 96
```

[22]: Text(0, 0.5, 'Coefficient magnitude')



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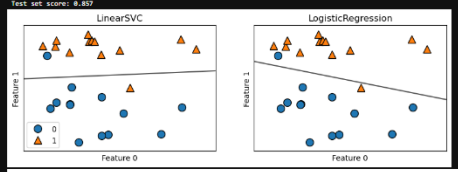
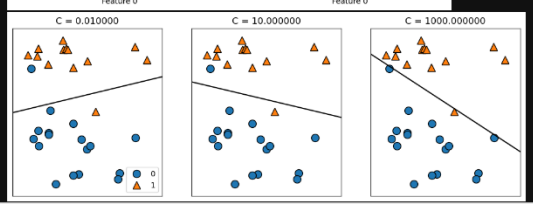
Code JupyterLab Python 3 (ipykernel)

[27]:
```python
import mglearn
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.model_selection import train_test_split

X, y = mglearn.datasets.make_forge()

fig, axes = plt.subplots(1, 2, figsize=(10, 3))

for model, ax in zip([LinearSVC(), LogisticRegression()], axes):
    clf = model.fit(X, y)
    mglearn.plots.plot_2d_separator(clf, X, fill=False, eps=0.5, ax=ax, alpha=.7)
    mglearn.discrete_scatter(X[:, 0], X[:, 1], y, ax=ax)
    ax.set_title("{}".format(clf.__class__.__name__))
    ax.set_xlabel("Feature 0")
    ax.set_ylabel("Feature 1")

axes[0].legend()

mglearn.plots.plot_linear_svc_regularization()

# split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)

logreg100 = LogisticRegression(C=100).fit(X_train, y_train)
print("Training set score: {:.3f}".format(logreg100.score(X_train, y_train)))
print("Test set score: {:.3f}".format(logreg100.score(X_test, y_test)))

logreg001 = LogisticRegression(C=0.01).fit(X_train, y_train)
print("Training set score: {:.3f}".format(logreg001.score(X_train, y_train)))
print("Test set score: {:.3f}".format(logreg001.score(X_test, y_test)))
```

```
Training set score: 1.000
Test set score: 0.857
Training set score: 0.947
Test set score: 0.857
```





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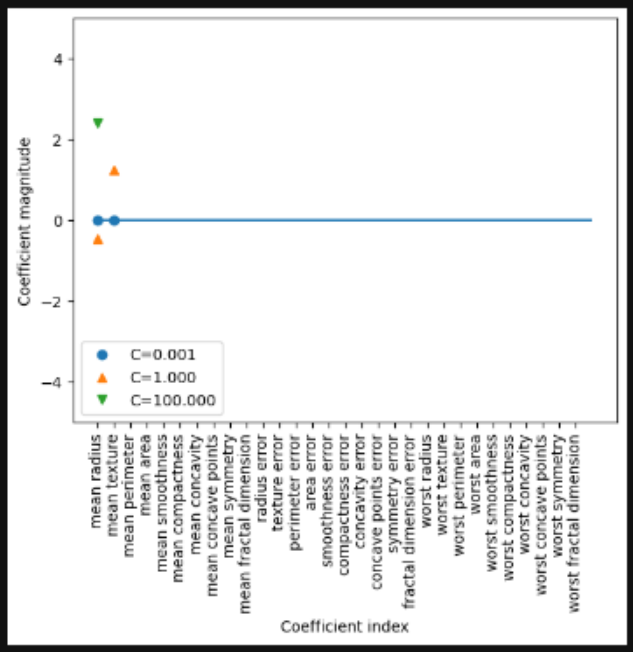
Code JupyterLab Python 3 (ipykernel)

[30]:
```
from sklearn.linear_model import LogisticRegression

for C, marker in zip([0.001, 1, 100], ['o', '^', 'v']):
    lr_l1 = LogisticRegression(C=C, penalty="l1", solver="liblinear", max_iter=5000).fit(X_train, y_train)

    print("Training accuracy of l1 logreg with C={:.3f}: {:.2f}".format(
        C, lr_l1.score(X_train, y_train)))

    print("Test accuracy of l1 logreg with C={:.3f}: {:.2f}".format(
        C, lr_l1.score(X_test, y_test)))

    plt.plot(lr_l1.coef_.T, marker, label="C={:.3f}".format(C))

plt.xticks(range(cancer.data.shape[1]), cancer.feature_names, rotation=90)
plt.hlines(0, 0, cancer.data.shape[1])
plt.xlabel("Coefficient index")
plt.ylabel("Coefficient magnitude")
plt.ylim(-5, 5)
plt.legend(loc=3)
```

```
Training accuracy of l1 logreg with C=0.001: 0.53
Test accuracy of l1 logreg with C=0.001: 0.43
Training accuracy of l1 logreg with C=1.000: 0.95
Test accuracy of l1 logreg with C=1.000: 0.86
Training accuracy of l1 logreg with C=100.000: 1.00
Test accuracy of l1 logreg with C=100.000: 0.86
```

[30]:
```
<matplotlib.legend.Legend at 0x29983a31d10>
```



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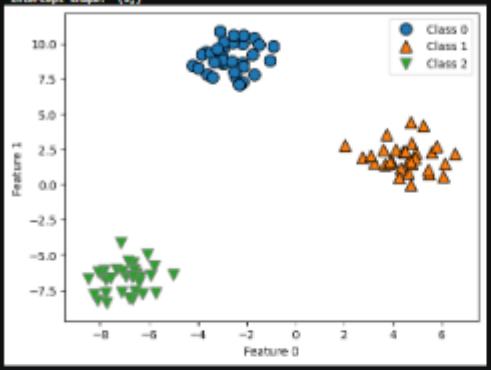
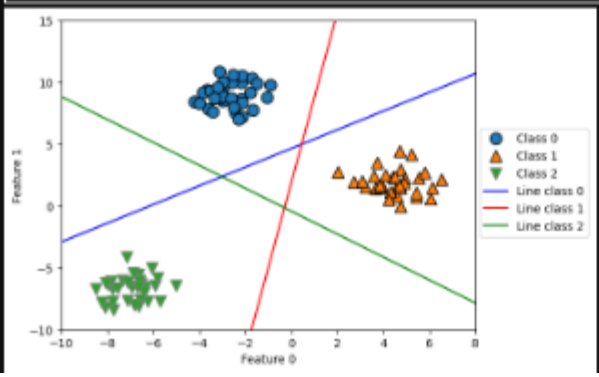
Code JupyterLab Python 3 (ipykernel)

[11]:
```
X, y = make_blobs(random_state=42)
mglearn.discrete_scatter(X[:, 0], X[:, 1], y)
plt.xlabel("Feature 0")
plt.ylabel("Feature 1")
plt.legend(["Class 0", "Class 1", "Class 2"])

linear_svm = LinearSVC().fit(X, y)
print("Coefficient shape: ", linear_svm.coef_.shape)
print("Intercept shape: ", linear_svm.intercept_.shape)
```

```
Coefficient shape:  (3, 2)
Intercept shape:  (3,)
```



[11]:
```
import numpy as np
import matplotlib.pyplot as plt
import mglearn
from sklearn.svm import LinearSVC

# create synthetic dataset (3 classes)
from sklearn.datasets import make_blobs
X, y = make_blobs(random_state=42)

# train linear SVM model
linear_svm = LinearSVC().fit(X, y)

# scatter plot of data points
mglearn.discrete_scatter(X[:, 0], X[:, 1], y)

# create line space
line = np.linspace(-15, 15)

# plot decision boundaries
for coef, intercept, color in zip(linear_svm.coef_, linear_svm.intercept_, ['b', 'r', 'g']):
    plt.plot(line, -(line * coef[0] + intercept) / coef[1], c=color)

# axis settings
plt.ylim(-10, 15)
plt.xlim(-10, 8)
plt.xlabel("Feature 0")
plt.ylabel("Feature 1")

# legend
plt.legend(['Class 0', 'Class 1', 'Class 2',
            'Line class 0', 'Line class 1', 'Line class 2',
           loc=(1.01, 0.3))

plt.show()
```



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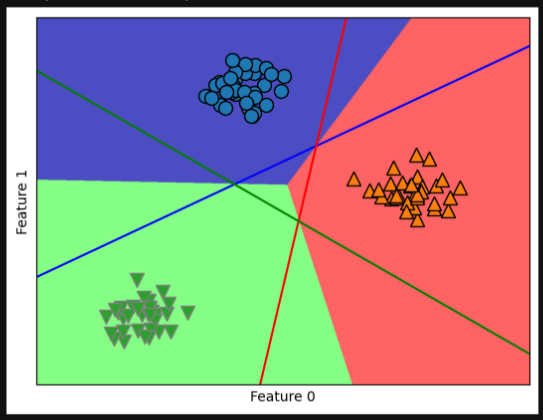
Code JupyterLab Python 3 (ipykernel)

[12]:
```
mglearn.plots.plot_2d_classification(linear_svm, X, fill=True, alpha=.7)
mglearn.discrete_scatter(X[:, 0], X[:, 1], y)
line = np.linspace(-15, 15)

for coef, intercept, color in zip(linear_svm.coef_, linear_svm.intercept_, ['b','r','g']):
    plt.plot(line, -(line * coef[0] + intercept) / coef[1], c=color)

plt.xlabel("Feature 0")
plt.ylabel("Feature 1")
```

[12]: Text(0, 0.5, 'Feature 1')



[ ]: